

A Randomized Trial of the Effect of Pharmacist Prescribing on Improving Blood Pressure in the Community: The Alberta Clinical Trial in Optimizing Hypertension (RxACTION)

Running title: *Tsuyuki et al.; Pharmacist prescribing in hypertension*

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Abstract

Background—Hypertension control rates remain suboptimal. Pharmacists' scope of practice is evolving and their position in the community may be ideal for improving hypertension care. We aimed to study the impact of pharmacist prescribing on blood pressure (BP) control in community-dwelling patients.

Methods and Results—We designed a patient-level randomized controlled trial, enrolling adults with above-target BP (as defined by Canadian guidelines), through community pharmacies, hospitals, or primary care teams in 23 communities in Alberta. Intervention group patients received an assessment of BP and cardiovascular risk, education on hypertension, prescribing of antihypertensive medications, laboratory monitoring, and monthly follow-up visits for 6 months (all by their pharmacist). Control group patients received a wallet card for BP recording, written hypertension information, and usual care from their pharmacist and physician. Primary outcome was the change in systolic BP at 6 months. A total of 248 patients (mean age 64 years, 49% male) were enrolled. Baseline mean (SD) systolic/diastolic BP was 150(14)/84(11) mm Hg. The intervention group had a mean (SE) reduction in systolic BP at 6 months of 18.3 (1.2) compared with 11.8 (1.9) in the control group, an adjusted difference of 6.6 (1.9) mmHg ($p=0.0006$). The adjusted odds of patients achieving BP targets was 2.32 (95% CI 1.17, 4.15 in favor of the intervention).

Conclusions—Pharmacist prescribing for patients with hypertension resulted in a clinically important and statistically significant reduction in BP. Policymakers should consider an expanded role for pharmacists, including prescribing, to address the burden of hypertension.

Clinical Trial Registration Information—clinicaltrials.gov; Identifier: NCT00878566.

Key words: hypertension, pharmacists, primary care

Introduction

Hypertension is a key risk factor for cardiovascular disease and premature mortality worldwide, affecting 1 in 5 North American adults and with 35-50% remaining uncontrolled.¹⁻² Since the prevalence of hypertension increases with age,²⁻³ it is expected that an aging population will lead to an even higher prevalence of hypertension and a greater burden on existing healthcare resources to manage elevated blood pressure (BP) and its sequelae in the coming years.

Given increasing workloads placed upon primary care physicians, pharmacists are well-positioned to take on a greater role in the management of chronic disease. Indeed, research has demonstrated the effectiveness of pharmacist-provided disease management activities, including for hypertension.⁴⁻¹⁰ Specific to hypertension, a recent systematic review and meta-analysis assessing the effect of pharmacist intervention for outpatients with hypertension identified 39 randomized controlled trials comprising over 14,000 patients.¹¹ Compared to usual care, interventions including pharmacists resulted in improvements in both systolic (-7.6 mm Hg, 95%CI: -9.0 to -6.3) and diastolic BP (-3.9 mm Hg, 95%CI: -5.1 to -2.8). Of note, a greater effect size was observed among the 23 studies where the pharmacist led the intervention, resulting in mean systolic and diastolic BP differences of -8.5 mm Hg and -4.6 mmHg, respectively, versus usual care. However, the pharmacist interventions provided largely involved patient education, medication management activities (defined as monitoring activities with medication adjustment), and recommendations to physicians. Therefore, the effectiveness of these interventions largely remain dependent on physician follow-through on drug therapy recommendations made by the pharmacists – a type of “ceiling effect”.

It is postulated that allowing pharmacists to independently prescribe drug therapy may result in even better patient outcomes than interventions based solely on providing

recommendations. Since 2007, the Health Professions Act of Alberta, Canada has allowed pharmacists to apply for authorization to prescribe.¹² To receive this authorization, pharmacists must have a minimum of one year of practice experience and must successfully complete an application process including the submission of patient care cases documenting their clinical involvement to demonstrate skills in patient assessment, judgment, care planning and follow-up.¹³ To ensure continuity of care, pharmacists must communicate all prescribing decisions to the patient's primary care physician and are responsible for conducting follow-up on their prescribing decision. Additionally, all Alberta pharmacists are able to order laboratory studies for patients under their care as required. Pharmacists are required to maintain professional liability insurance.

The current study was designed to test the hypothesis that pharmacist prescribing for community-dwelling patients with uncontrolled hypertension would result in improved blood pressure reduction over usual care.

Methods

The design and methods of this study have been published previously and are outlined in **Figure 1**.¹⁴ Briefly, we conducted a multicenter randomized controlled trial comparing enhanced pharmacist care (which included independent patient assessment, counseling and prescribing) with usual care in the province of Alberta, Canada. All participating pharmacists had prescribing authorization and practiced in community (20 pharmacists), hospital outpatient clinics (2 pharmacists), or primary care clinic settings (6 pharmacists). Pharmacists received training in blood pressure assessment and treatment based on the Canadian Hypertension Education Program (CHEP) guidelines¹⁵, and had access to hypertension experts for consultation as required.

Eligible patients were identified opportunistically through BP screening events, case finding¹⁶ by identifying patients at high risk of elevated BP, and through the course of providing routine care. Patients were enrolled into the study by participating pharmacists, but could also be referred for assessment regarding study eligibility by a patient's physician, nurse, or other health professionals. Patients were eligible for the study if they were adult outpatients and had uncontrolled BP as defined by the CHEP guidelines (which mandated multiple visits to define uncontrolled BP, with the exact number depending on the level of BP). Overall, this meant a BP >140/90 mmHg for most, and >130/80 mmHg for those with diabetes. As the CHEP guidelines specify a number of diagnostic scenarios with different BP cut-offs and number of readings required, readers are referred to the guidelines for the clinical definitions of 'uncontrolled BP' for the purpose of this study.^{15,17-20} Patients meeting any of the criteria for diagnosis or, if already diagnosed, remaining above the specified target given their age and comorbidities, could be included. Any changes to these guideline definitions over the course of the study were concurrently applied to the inclusion criteria for the study.

Exclusion criteria included hypertensive urgency or emergency (defined as systolic BP ≥ 200 and/or diastolic BP ≥ 130 mm Hg with/without symptoms), pregnancy, or patients unwilling or unable to provide consent to participate. The study received ethics approval from the University of Alberta Health Research Ethics Board, and written informed consent was obtained from all participants.

All BP measurements performed by the pharmacist were made using the automated BpTRU[®] (BpTRU Medical Devices, Coquitlam, British Columbia, Canada), which takes 6 readings, discarding the first and takes the average of the remainder. Home measurements were performed using the LifeSource UA-787 home blood pressure monitor. Both devices are

validated and/or approved by CHEP, and pharmacists were to ensure that proper cuff sizes were used for all measurements.^{21,22} Home measurement was employed for those patients requiring more than 2 office screening visits, as defined by the CHEP guidelines,^{15,17-20} as a guideline-concordant option for obtaining these additional BP readings. All patients utilizing home measurement were provided verbal and written instruction on proper measurement technique by the pharmacist, and were provided a cuff of proper size to ensure accuracy.

Randomization was conducted at the level of the patient, and was performed via a centralized secure website at the Epidemiology Coordinating and Research (EPICORE) Centre (<http://www.epicore.ualberta.ca>) to ensure randomization concealment. Patients were randomized in a 2:1 ratio to either intervention or usual care. Due to the nature of the intervention, blinding was not possible.

The intervention group received “enhanced” pharmacist care, which was guided by the CHEP guidelines and consisted of pharmacist assessment of, and counseling about cardiovascular risk and BP control, review of antihypertensive medications, and prescribing/titrating of drug therapy if deemed necessary, in addition to a wallet card for recording BP measurements, lifestyle advice, and written information on hypertension developed by CHEP. The patient’s primary care physician was notified of all assessment results and drug therapy changes in person or by fax. Intervention group patients were followed up at monthly intervals until their BP was at target for two consecutive visits and, thereafter, at 3-month intervals for the duration of the study period as per CHEP recommendations.

The usual care group received a wallet card for recording BP, lifestyle advice as required, written information on cardiovascular disease, and BP measurement by the pharmacist at 3-month intervals. Patient education was provided the discretion of each pharmacist. Patients’

primary care physicians were sent a notice that the patient was enrolled in the study based on their elevated BP, and patients were advised to see their physician for further treatment. All patients were followed for a total of 6 months.

The primary outcome was the difference in change in systolic BP from baseline to 6 months between intervention and usual care groups. Secondary outcomes include the change in diastolic BP and the number of patients at their CHEP-recommended target BP (defined as both systolic and diastolic BP at target, i.e., <130/80 mm Hg for those with diabetes and <140/90 mm Hg for all others) after 6 months and, in the intervention group, the number of new antihypertensive medication starts, the number of antihypertensive dose changes, the number of antihypertensive drug changes, and the number of new prescriptions for ASA and cholesterol lowering medications. Drug therapy changes were enumerated at the level of the drug therapy class; in other words, an individual patient with two new antihypertensive drug classes initiated would be counted as two new starts. All outcomes were measured by the pharmacist investigators in the process of providing care to study patients.

Since the initial publication of the study's methods, some protocol modifications were made to remain consistent with updated hypertension guidelines and to enhance participant recruitment. While the study was initially limited to rural communities, the enrollment criteria were revised to include urban centers in January 2012. Additionally, reflecting changes in the CHEP recommendations in 2012, undiagnosed and uncomplicated patients with an average BP across 2 visits of 140-179/90-109 mm Hg could be further screened through 24-hour ambulatory blood pressure monitoring. Patients could be enrolled in the study with an average 24-hour BP of systolic ≥ 130 mm Hg or diastolic ≥ 80 mm Hg, or average awake-hours BP of systolic ≥ 135 mm Hg or diastolic ≥ 85 mm Hg. Target BP for patients with chronic kidney disease in the absence of

diabetes also changed from <130/80 mm Hg to <140/90 mm Hg in the 2012 CHEP guidelines¹⁷ and was incorporated into study targets from that point onward.

As outlined in our methods publication¹⁴, our pre-specified sample size was 340 patients, including 90 in the usual care arm and 250 in the enhanced pharmacist care arm. All analyses were conducted using IBM SPSS Statistics, version 21 (IBM Corp., Armonk, NY) and SAS, version 9.4 (SAS Institute Inc., Cary, NC) and followed the intent-to-treat principle, with *P* set at 0.05. A generalized multivariable linear regression with change in systolic BP and diastolic BP as the dependent variables was performed to adjust for baseline imbalances between groups (defined as those characteristics with $p < 0.20$). As such, we adjusted for a history of myocardial infarction, presence of diabetes, and a first degree relative with a history of stroke. Similarly, a logistic model was also used to obtain an adjusted odds ratio for those achieving the target BP. To account for within pharmacist correlation, all models used robust estimates of variance, clustered by every pharmacist who recruited patients into the study. Missing values were imputed using the last-observation carried forward method.

Results

Between July 2009 and May 2013, investigators at 23 pharmacies screened 754 patients and enrolled 248 (181 randomized to intervention, and 67 to usual care, Figure 2). Enrollment was stopped prior to accrual of the full sample size due to funding limitations. Of the 506 patients who were not randomized, 495 were ineligible on the basis of not having elevated BP as defined by CHEP at the time of measurement^{15,17-20}, 10 did not provide consent, and 1 was referred for immediate care of hypertensive urgency/emergency. A total of 32 patients withdrew from the study (26 (14%) in the intervention group and 6 (9%) in the usual care group). Patients who

withdrew from the study generally did not differ from those who continued in the trial, with the exceptions of being older (mean 68 years vs 62, $p=0.02$) and living a greater distance from the pharmacy (mean 25 kilometers vs 12, $p<0.001$). Completeness of follow-up was 85% in the intervention group, and 91% in the usual care group.

Intervention and usual care groups were similar at baseline (**Table 1**). Patients' average (SD) age was 64 (12) years and 49% were male. The overall mean (SD) systolic/diastolic BP was 150(14) / 84(11) mm Hg at baseline, and the majority (78%) were already taking antihypertensive drug therapy, taking an average of 1.7 medications. Patients already receiving antihypertensive therapy at baseline were older (mean 65 vs. 58 years, $p=0.001$), had lower diastolic BP (mean 82.6 vs. 86.6 mm Hg, $p=0.02$), were more likely to have a history of MI ($p=0.02$), atrial fibrillation ($p=0.05$), stroke ($p=0.02$), peripheral artery disease ($p=0.03$), and family history of MI ($p=0.04$), and less likely to have a history of dyslipidemia ($p=0.04$) than those not on drug therapy. Of those not receiving antihypertensive therapy at enrolment, nearly half (49%) had been previously diagnosed with hypertension.

Systolic BP decreased in both groups over the 6-month trial, with a greater reduction observed in the intervention group (**Figure 3**). After adjustment, the mean (SE) difference in systolic BP between groups was 6.6 (1.9) mm Hg, which was statistically significant ($p=0.0006$). Diastolic BP also decreased in both groups, with an adjusted mean (SE) difference of 3.2 (1.3) mm Hg, $p=0.01$.

The proportion of patients achieving Canadian Hypertension Education Program - recommended target BP was also significantly higher in the intervention versus the usual care group (crude rates of 58% intervention vs. 37% usual care, $p = 0.02$) with an adjusted odds ratio of 2.32, 95% CI 1.17, 4.15.

Antihypertensive medication use for each group is provided in Table 2. In the intervention group (n=181), 103 new antihypertensive drugs were initiated, 94 dose changes made (80 dose increases, 14 dose decreases), 76 antihypertensive drugs were discontinued. Also, 12 patients were prescribed low-dose ASA and 14 prescribed a statin by the pharmacist. In the usual care group (n=67), 20 new antihypertensive drugs were initiated, 9 dose changes made (8 dose increases, 1 dose decreases), 15 antihypertensive drugs were discontinued, 2 patients were initiated on low-dose ASA, and 2 on a statin.

Discussion

An expanding scope of practice for pharmacists is increasingly being adopted worldwide for a number of reasons including pharmacists' accessibility and drug therapy expertise and the increasing prevalence of chronic diseases, which are largely managed through lifestyle interventions and pharmacotherapy. To our knowledge, our study is the first randomized trial of independent pharmacist prescribing in community-dwelling patients with hypertension, and demonstrated significant reductions in systolic and diastolic BP, and the proportion of patients reaching recommended BP targets, compared to usual pharmacist and physician care. Given the accessibility of community pharmacies, and high burden of illness from hypertension, these findings could have important public health impact.

An innovation of our study was the independent prescribing authorization of pharmacists. We hypothesized that pharmacist prescribing could overcome the ceiling effect that may be associated with recommendation-based care, and we expected that the difference in change in systolic BP from this intervention would be even greater than the 6.6 mm Hg observed. However, we observed a higher than expected BP reduction among the usual care group of 11.8

mm Hg, more than double that observed in the control arm of the *SCRIP-HTN* study in a similar population.²³ Potential contributing factors could be the Hawthorne effect among patients and their primary care physicians, the effectiveness of the CHEP program's knowledge translation efforts to improve awareness and control of hypertension in Canada,²⁴ case finding efforts by pharmacists which may have preferentially enrolled patients with other comorbidities and therefore potentially differing motivation than identified via general population screening,¹⁶ or the provision of interventions other than prescribing by the pharmacist which may have resulted in improved blood pressure management. Since we relayed BP measurements to the patient and their family physician in the usual care group (thus functioning as an audit and feedback intervention in our "usual care" arm, which actually represents more than usual care), this may have also improved their blood pressure care. Nevertheless, all such contaminating factors would have biased our findings towards the null hypothesis. Additionally, it should be noted that a slightly greater proportion of intervention patients were lost to follow-up than usual care patients, however we used a last value carried forward approach to impute missing values, which would have also biased towards the null hypothesis.

In comparison to the results of a very comprehensive recent meta-analysis of 39 randomized trials examining the effectiveness of pharmacist interventions by Santschi and colleagues,¹¹ our study sample was similar in terms of participants' age and sex, and frequency of intervention visits, but had a slightly shorter intervention duration (6 months vs. mean of 8.2 months). We found slightly lower mean differences in both systolic BP (-6.6 vs -7.6 mm Hg) and diastolic BP (-3.2 vs -3.9 mm Hg); however, as mentioned above, the marked improvements noted among the usual care group from potential contamination offset the absolute BP reductions observed in the intervention group of 18.3 and 8.0 mm Hg for systolic and diastolic BP,

respectively. In contrast to the Santschi meta analysis¹¹, our study involved primarily community pharmacies, which may provide more generalizability in terms of potential public health impact.

It is also important to note that pharmacists providing care for intervention group patients were also remunerated for their services as part of the study. As described in the study protocol,¹⁴ this remuneration was part of a secondary randomization process to function as a sub-study of the main analysis (which will be reported separately). In the substudy, each patient randomized to the intervention group was further randomized in a 1:1 ratio to pharmacist payment in the form of fee-for-service (a flat rate per visit regardless of outcome) or pay-for-performance (a flat rate per visit with incentive payments for achieving BP goals). One cannot ignore the potential impact of this arrangement on the pharmacists choosing to participate in the study, the number of patients enrolled, or the nature of the care provided. As of July 2012, Alberta pharmacists have been able to claim a fee from the Alberta government for medication management services.²⁵ Pharmacists enrolling patients into the study prior to this date would therefore have received remuneration for providing care for study patients that otherwise would not have been offered. Conversely, after July 2012, this fee discrepancy would have been reduced, as fee-for-service remuneration would have been available for these patients outside of the study environment.

This study is not without limitations. Due to the nature of the intervention, neither the patients nor the pharmacist investigators could be blinded to the treatment group to which patients were allocated. However, our outcomes were objective (measured BP) and captured by automated devices, removing any potential for differential measurement bias between arms. Additionally, by design, patients were seen more frequently by the pharmacist if randomized to intervention versus usual care, which may have contributed to the blood pressure reduction observed due to reduced white coat hypertension over time or regression to the mean. To

minimize the impact of white coat hypertension, a well-validated automated blood pressure device (BpTRU[®]) was used for all study measurements. Pharmacists were also asked to vacate the room when these measurements were taken to further reduce patient anxiety. Additionally, with approximately 10% of Alberta pharmacists currently possessing additional prescribing authorization (Alberta College of Pharmacists, personal communication), it is possible that the early adopters choosing to apply for additional prescribing authorization and to participate in this study may differ from the ‘typical’ pharmacist,²⁶ affecting the generalizability of the results. Similarly, one cannot assume that those patients volunteering for the study are representative of the population in terms of health motivation, potentially explaining some of the BP reduction noticed even among usual care patients. One also must consider the possibility for contamination among the usual care group since we did not employ cluster randomization (due to logistics and the small number of pharmacists with additional prescribing authorization when the study was launched) and, thus, the usual care provided by pharmacists in our study may differ from the usual care provided by all pharmacists. Furthermore, while most drug therapy changes in the intervention group were likely implemented by the pharmacist, one cannot rule out that some medication adjustments were performed by the patient’s primary care physician. However, the small number of drug changes in the usual care arm would argue against this being a major confounder. Finally, we did not achieve our target sample size of 340 patients and due to funding limitations, terminating the enrolment at 248 patients. It should be noted, however, that the sample size of 340 was required for the remuneration substudy (our *a priori* sample size calculation for the main study was 240 patients).

The results of our study demonstrate that pharmacist prescribing, when provided in addition to usual care, results in a clinically significant reduction in BP and a substantial

improvement in the proportion of patients with initially uncontrolled hypertension reaching their target BP even though a very high proportion, 78%, were already taking antihypertensive therapy at baseline.

While Alberta was the first Canadian jurisdiction allowing pharmacists to independently prescribe antihypertensive drug therapy for patients, the provinces of Manitoba and New Brunswick, Canada, has recently adopted similar legislation for pharmacists with specialty training and working as part of a collaborative practice,²⁷ and independent pharmacist prescribing has been in place in the United Kingdom since 2006.²⁸ The generalizability of our study results to these, and future, regions utilizing independent pharmacist prescribing is unknown, and ongoing evaluation of these programs is encouraged. This study, the first randomized controlled trial of pharmacist prescribing, and prodigious evidence from 39 non-prescribing trials¹¹ supports recent efforts to expand pharmacists' scope of practice to include medication management activities in an effort to address clinical inertia in hypertension management.²⁹⁻³¹

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Table 1. Patient Characteristics.

Variable	Intervention (n=181)	Usual Care (n=67)
Demographics:		
Male sex	89 (49)	32 (48)
Age, mean (SD) in years	63 (13)	65 (11)
Urban residence	103 (57)	33 (49)
Cardiovascular risk factors:		
Systolic BP at baseline, mean (SD) in mm Hg	150 (15)	151 (11)
Diastolic BP at baseline, mean (SD) in mm Hg	85 (12)	83 (10)
1 st degree relative history of MI	87 (48)	32 (48)
1 st degree relative history of angina	47 (26)	20 (30)
1 st degree relative history of stroke	57 (32)	27 (40)*
BMI, mean (SD)	32 (7)	32 (7)
Waist circumference, mean (SD) in cm	106 (17)	109 (14)
Elevated waist circumference (>102 cm in men, >88 cm in women)	126 (70)	51 (76)
Smoking		
Current	32 (18)	9 (13)
Ex-smoker	78 (43)	28 (42)
Never	68 (38)	30 (45)
Alcohol consumption		
One or more servings per day	28 (16)	8 (12)
Occasional	90 (50)	37 (55)
Salt added to food		
Often/always	31 (17)	11 (16)
Sometimes	41 (23)	22 (33)
Self-reported cardiovascular comorbidities:		
Diabetes	71 (39)	38 (57)*
Chronic kidney disease	31 (17)	7 (10)
History of MI	8 (4)	8 (12)*
History of angina	23 (13)	13 (19)
History of heart failure	2 (1)	0
History of atrial fibrillation	22 (12)	5 (8)
History of stroke	10 (6)	6 (9)
Dyslipidemia	93 (51)	37 (55)
Peripheral artery disease	11 (6)	4 (6)
Prior revascularization procedure	11 (6)	6 (9)
On antihypertensive drug therapy at baseline	137 (76)	56 (84)
Number (SD) of drugs taken	1.7 (1.2)	1.7 (1.2)

Abbreviations: SD, standard deviation; MI, myocardial infarction; BMI, body mass index (calculated as weight in kilograms divided by height in meters squared).

All data are given as numbers (percentages) unless otherwise indicated.

* Characteristics with baseline differences between groups at $p < 0.20$ and, therefore, included in multivariable models.

Table 2. Use of antihypertensive medications.

	Intervention, No. (%) (n=181)		Usual Care, No. (%) (n=67)	
	Baseline	6 Months	Baseline	6 Months
Thiazide diuretic	77 (43)	99 (55)	25 (37)	30 (45)
ACE inhibitor	65 (36)	68 (38)	29 (43)	26 (39)
Beta blocker	34 (19)	35 (19)	9 (13)	10 (15)
Calcium channel blocker	52 (29)	67 (37)	21 (31)	22 (33)
Angiotensin receptor blocker	63 (35)	76 (42)	25 (37)	29 (43)
Other	13 (7)	12 (7)	5 (8)	6 (9)

Abbreviation: ACE, angiotensin-converting enzyme

Figure Legends:**Figure 1.** Study Design.**Figure 2.** Trial Flow.**Figure 3.** Adjusted* difference in systolic and diastolic blood pressure (BP) over 6 months.

*Adjusted for history of myocardial infarction, diabetes and first degree relative with stroke.

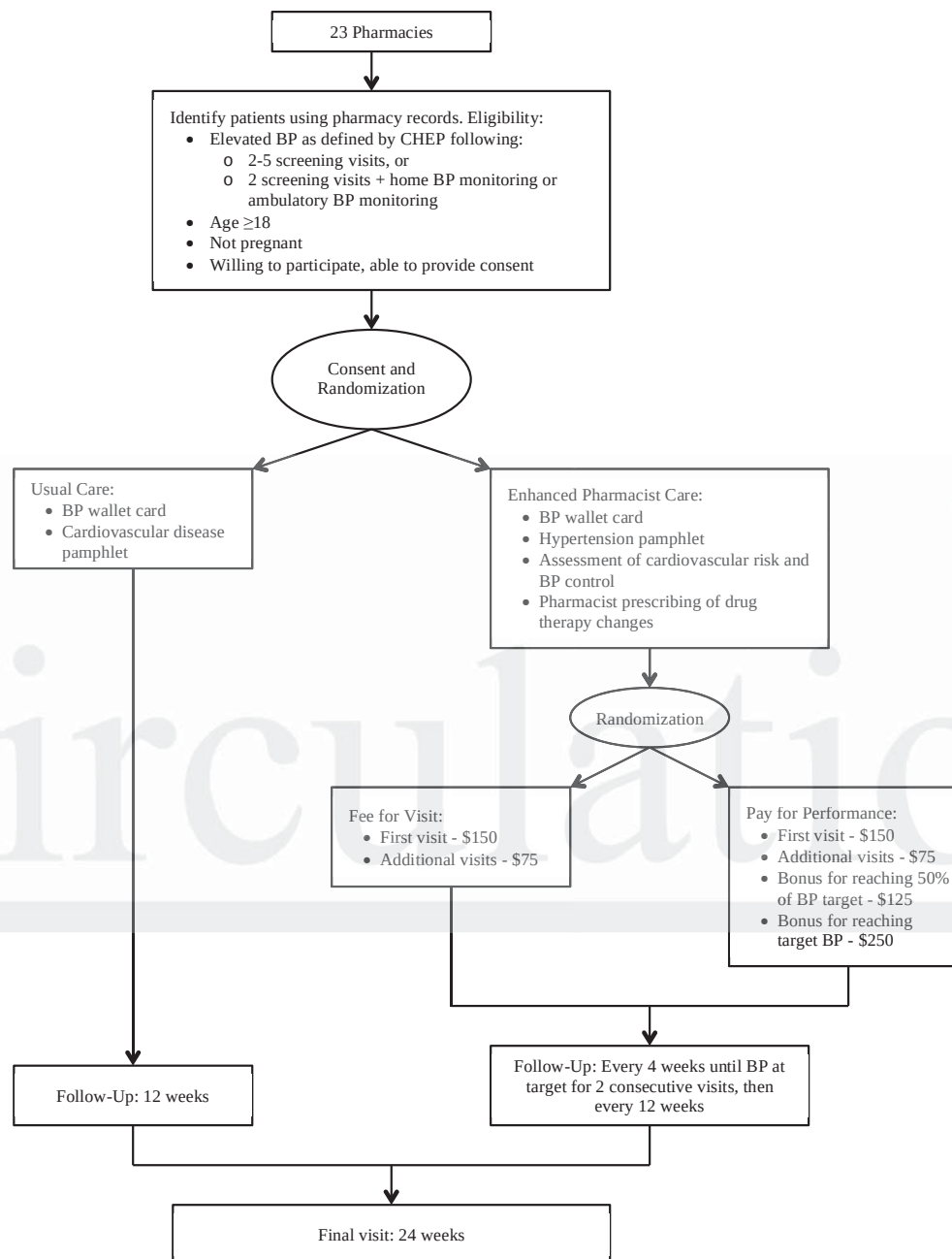


Figure 1

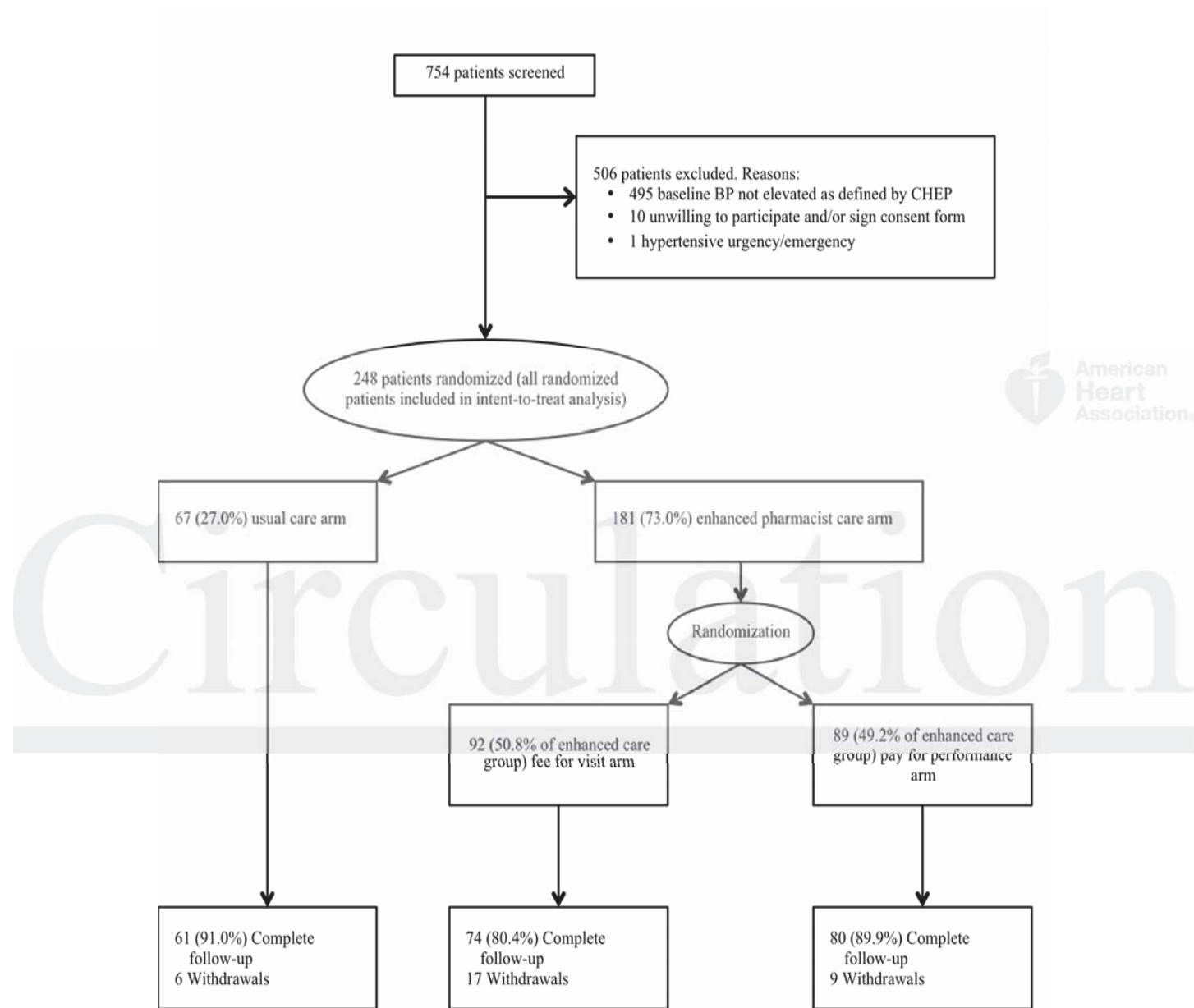


Figure 2

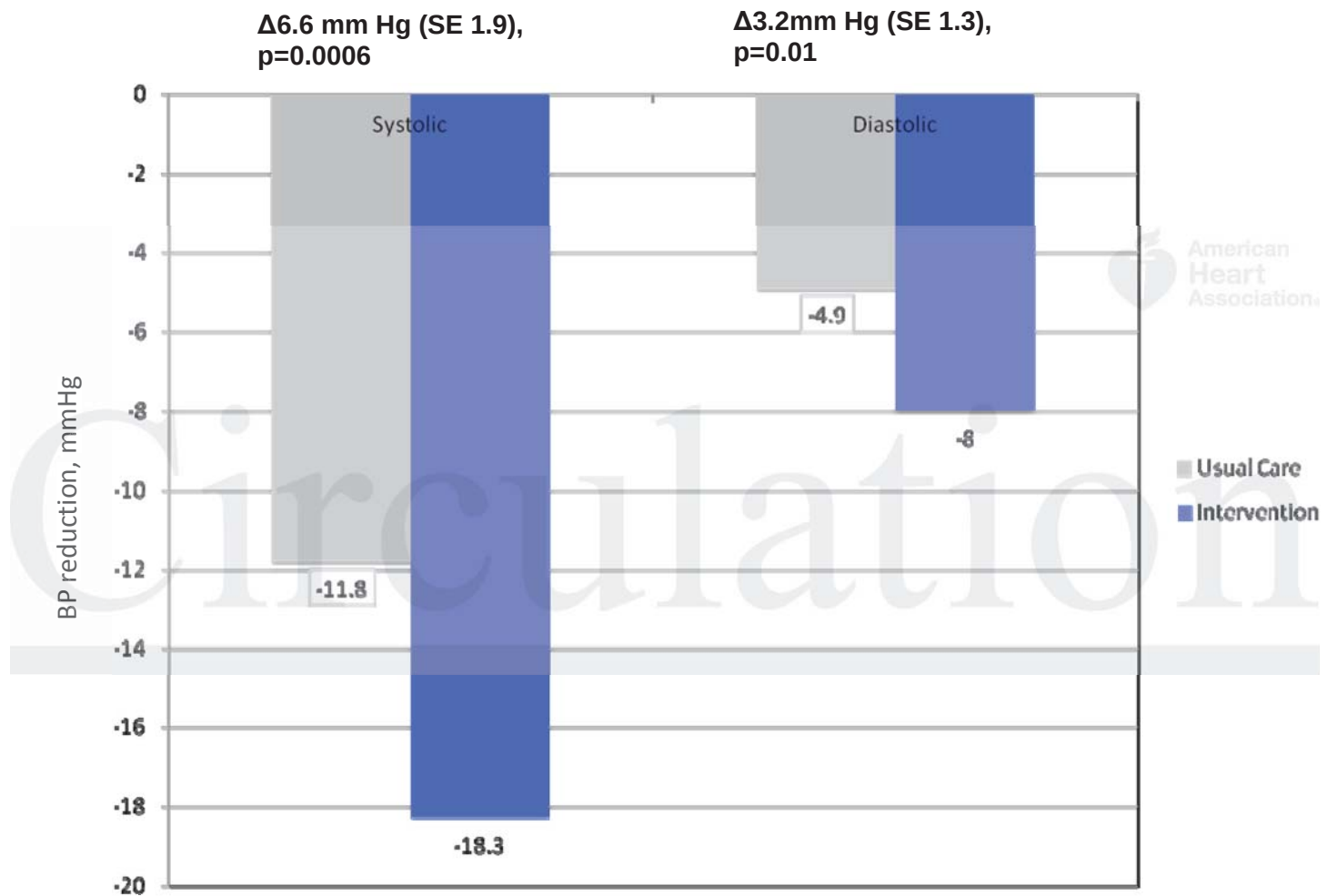


Figure 3

SUPPLEMENTAL MATERIAL

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A Randomized Trial of the Effect of Pharmacist Prescribing on Improving Blood Pressure in the Community: The Alberta Clinical Trial in Optimizing Hypertension (RxACTION)

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